

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	xxxxxxx
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform how undergraduate engineering and language students approach dense learning materials using generative artificial intelligence. It tackles academic information overload, promising better document parsing, customized progress analytics, and more confident exam preparation. Key features include an AI-driven multi-modal contextual engine and real-time interactive assessment generation.

Project Overview	
Objective	The primary objective is to revolutionize the self-paced study process by implementing advanced cloud and local AI techniques, ensuring faster, highly targeted, and more accurate conceptual comprehension assessments.
Scope	The project comprehensively assesses and enhances custom document ingestion pipelines, incorporating localized explanation engines and real-time evaluation frameworks for a more robust and responsive digital learning dashboard.
Problem Statement	
Description	Addressing the mass inaccuracies and cognitive inefficiencies in trying to manually parse long-form academic material, dense textbooks, and complex terminology, which adversely affects student comprehension speeds and performance scores.

Impact	Solving these issues will result in improved personal study efficiency, reduced anxiety during crunch periods, and an overall enhancement in visual metrics tracking, directly contributing to student success and academic excellence.
Proposed Solution	
Approach	Employing machine learning and cloud generative AI techniques to analyze, simplify, and extract meaning from uploaded text, creating a dynamic, interactive, and adaptable dashboard environment.
Key Features	- Implementation of a cloud-based generative AI core metrics assessment module for customized learning roadmaps. - Real-time automated test generation (quizzes and performance scoring engines) for instantaneous evaluation metrics. - Continuous tracking and local explainers to adapt to dense, evolving academic domains or technical vocabularies.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Cloud & Local inference allocations	T4 GPU or optimized local hardware
Memory	RAM specifications	8 GB
Storage	Disk space for parsing libraries, state data, and temporary logs	1 TB SSD
Software		
Frameworks	Web interface implementation backend	Python (Flask / FastAPI)
Libraries	Additional computation & modeling tools	scikit-learn, pandas, numpy, matplotlib, seaborn

Development Environment	Primary deployment & building workspaces	Jupyter Notebook, PyCharm, VS Code
Data		
Data	Parsing structures, size, format	Custom uploaded text files (PDF, TXT) / Academic datasets